Cost per Kilogram — DAO 2023 vs DAO 2026 vs DHL vs FedEx

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1. Headline: cost per kilogram (USD)

Option	Weight basis	FREIGHT-ONLY $/kg	ALL-IN $/kg
DAO 2023 actual (Seacoast + Omega)	104 kg	**7.46**	**14.09**
DAO 2026 actual (5 Continent, +confirmed)	349 kg	**3.56**	**14.34**
DHL Express — published retail, heavy band	68-150 kg	18.74 - 84.80	n/a (freight only)
DHL Express — typical discounted / agent (46-100 kg)	46-100 kg	5.00 - 6.80	n/a (freight only)
FedEx International Priority Freight — published, 100-499 kg	100-499 kg	7.90 - 21.90 (by zone)	n/a (freight only)

Bases matter — read this first:

- The **DAO** figures are **ALL-IN landed cost** (air freight + inland + customs + FDA + duties + terminal + delivery). Its **freight-only** slice is the air leg alone (Omega numerario in 2023, the air-freight line in 2026).
- **DHL / FedEx** figures above are **freight-only** (transport). Customs, duties, and clearance are billed separately / on account, so they are NOT in these numbers.
- So the correct like-for-like on **freight-only** is: DAO 2023 $7.46, DAO 2026 $3.56, vs DHL $5-85, FedEx $8-22.

2. Like-for-like: FREIGHT-ONLY comparison

Option	Weight	Freight-only $/kg
DAO 2026 actual	349 kg	**3.56**
DAO 2023 actual	104 kg	**7.46**
DHL typical discounted / agent	46-100 kg	5.00 - 6.80
FedEx IPF published (low zone)	100-499 kg	7.90
FedEx IPF published (high zone)	100-499 kg	20.30 - 21.90
DHL published retail (low zone)	68-150 kg	18.74
DHL published retail (high zone)	68-150 kg	84.80

Read: the DAO's forwarder lane (freight-only $3.56-$7.46/kg) sits **at or below the cheapest band** of every express courier. Express couriers are competitive only at their deepest discount tiers and lowest zones; at published retail on a long-haul lane they run **5x-10x** the DAO's freight rate.

3. ALL-IN comparison (DAO all-in vs carrier freight-only, clearly separated)

Option	Basis	$/kg
DAO 2023	all-in landed	14.09
DAO 2026 (+confirmed)	all-in landed	14.34
DHL Express	freight only, retail	18.74 - 84.80
FedEx IPF	freight only, retail	7.90 - 21.90

- Even before customs is added, **DHL retail freight alone exceeds the DAO's full landed cost** on most zones. FedEx freight alone straddles it.
- Adding customs/duties/FDA to a DHL or FedEx quote would push the carrier all-in comfortably **above** the DAO's ~$14/kg.

4. Basis & sources

Figure	Source
DAO 2023 (104 kg)	2023 reconciliation: Seacoast invoice #102000607-1 (US side) + Omega numerario R$4,000 + CT-e R$864.15
DAO 2026 (349 kg)	5 Continent invoice #1026-08 + cintagem R$300 + fumigation R$405, at BACEN PTAX 5.1575
DAO freight-only slice	air leg only: R$4,000 (2023) / $1,241 (2026)
DHL published retail	DHL Express Service & Rate Guide 2026 (US), per-lb multiplier 150.1-330 lb band: $8.50-$38.48/lb
DHL discounted / agent	Tonlexing reference rate card, 46-100 kg band (China-EU lane)
FedEx published	FedEx International Priority Freight 2026 rate card, per-kg by zone, 71-1,000 kg tiers
FX	EUR->USD approx 1.08; BRL->USD at DAO ledger 0.19405 (2023) / PTAX 5.1575 (2026)

5. Caveats

- **DHL / FedEx are published list rates.** Business accounts routinely get **15-40% off** (DHL 2026 notes dynamic discounts start at 40% off retail); heavy-weight tiers are **quote-only**. A real quote needs exact dims + origin/destination postcodes.

• **Zone dominates carrier pricing.** The DHL/FedEx spreads above are *across zones*, not uncertainty — Brazil-to-US long-haul sits in the higher zones, so expect the upper half of each range.

• **SeaCoast forwarder estimate** (your "Costing estimates" sheet, all-in SUBTOTAL) for reference: $11.81/kg @ 200 kg, ~$8.12/kg @ 349 kg, $4.91/kg @ 1,000 kg — quoted on a $1,000 cargo-value placeholder, so duty/bond would lift it at real value.

• **DAO 2026 cargo is under the arrival gate** — nothing here touches balances or lands inventory.

6. Bottom line

• **Freight-only:** DAO 2026 ($3.56/kg) is the cheapest; DAO 2023 ($7.46/kg) is mid; DHL discounted and FedEx low-zone are comparable; DHL published retail is far higher.

• **All-in:** the DAO's ~$14/kg landed is **below DHL published freight alone** and around the FedEx freight floor — before any customs is added to the carrier side.

• For **bulk cacao at 100-350 kg**, the DAO's forwarder lane is the economic choice; express couriers only make sense for urgent, small, high-value parcels.